MIL-S-37800A
24 August 1978
SUPERSEDING
MIL-S-37800(DLA-DM)
31 October 1977

MILITARY SPECIFICATION

SUNBURN PREVENTIVE PREPARATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sunburn Preventive Preparation (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fibreboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-105 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6508

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a sunscreen lotion containing the following formula:

Ingredients	W/W Percent
Para-aminobenzoic acid	5
Ethyl Alcohol	50 to 55 (Formula Weight)
Other suitable ingredients to prepare a lotion 1/	qs 100

1/ May consist of emollients, thickening agents, preservatives, etc., acceptable to FDA.

3.1.1 Assay.

3.1.1.1 Para-aminobenzoic acid. The lotion shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of para-aminobenzoic acid when determined as specified in 4.7.1.

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3.1.1.2 Alcohol. The lotion shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the labeled amount of alcohol which shall be between 50 and 55 percent, when determined using USP alcohol content method.

3.1.2 Identification. The presence of para-aminobenzoic acid in the lotion shall be determined as specified in 4.7.2.

3.2 Ingredients.

3.2.1 Aminobenzoic acid and alcohol. The aminobenzoic acid and alcohol used in the manufacture of the lotion shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the lotion shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade suitable for use in the lotion.

3.4 Quantity of contents. Immediate container (bottle) may be labeled from 118 ml (4 fl oz) to 120 ml.

3.5 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the lotion to the date of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of product produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Units of product
Major	S-2	1.0	Filled bottle
Minor	II	2.5	Filled bottle

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection. The acceptance number for each test shall be zero (0).

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay (Para-aminobenzoic acid)	3.1.1.1	S-2	5 } Composite
Assay (alcohol)	3.1.1.2	S-2	5 } Composite
Identification	3.1.2	S-2	5 } Composite

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished lotion shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Lotion not essentially free from foreign matter.
102	Lotion not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

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4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be of the unit of issue. The inspection lot shall be the numbe of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for para-aminobenzoic acid.

The samples shall be assayed as directed in the USP monograph for Aminobenzoic Acid Solution. As an alternate, the samples may be assayed by the following method.

Prepare a standard solution containing 5 mcg of para-aminobenzoic acid per ml in ethanol or isopropyl alcohol. Note: The para-aminobenzoic acid standard shall be tested against the requirements in the monograph for Aminobenzoic Acid USP and shall meet all requirements of the monograph. Test records delineating the results shall be made available, if requested. Test comparison for each lot of standard with Aminobenzoic Acid USP need be performed one time only.

Procedure.

Weigh accurately a portion of sample equivalent to 25 mg of para-aminobenzoic acid and transfer quantitatively into a 100 ml volumetric flask with 70 ml of ethanol and shake well. Dilute to 100 ml mark with ethanol and filter solution if not clear. Pipet 2 ml of this solution to another 100 ml volumetric flask and add sufficient quantity of ethanol or isopropyl alcohol to make up to volume. Shake well and concomitantly determine the absorbances of sample and standard solutions in 1 cm cells at the wavelength of maximum absorbance at about 287 nm to 289 nm with a suitable recording spectrophotometer, using ethanol or isopropyl alcohol as the blank.

Calculation

$$\text{Percent Para-aminobenzoic Acid} = \frac{Au}{As} \times \frac{Cs}{Sw} \times 500$$

Where:

Au = Absorbance of sample.
As = Absorbance of standard.
Cs = Concentration of standard (gram/liter)
Sw = Sample weight (gram).

NOTE: As an alternate, SD Alcohol 3A may be used as the solvent in lieu of ethanol or isopropyl alcohol. In addition the standard and sample weights (quantities) and their dilutions may vary provided their final concentration is approximately 5 mcg per ml.

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4.7.2 Identification of para-aminobenzoic acid.

a. The ultraviolet absorption spectrum obtained in the assay for para-aminobenzoic acid shall exhibit maxima and minima at the same wavelengths as that of a similar preparation of para-aminobenzoic acid standard.

b. Shall comply with the identification tests A and B in the USP monograph for Aminobenzoic Acid Solution.

Note: If the UV assay for para-aminobenzoic acid is followed, then identification test (b) need not be performed.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing twelve bottles, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Three (3) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Three (3) units of issue shall be packed in an exterior container conforming to PPP-B-636, type CF, class domestic.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

5.4.2 Exterior container. Exterior container shall be marked in accordance with 5.5.4 of PPP-C-186, except that the date of manufacture shall be shown in lieu of the date packed.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification
6508-01-037-8636	SUNBURN PREVENTIVE PREPARATION, 120 ml, 12s or 118 ml, 12s 1/ Firm may use either of the above designations (see 3.4).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6508-0007

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MIL-37800A SUNBURN PREVENTIVE PREPARATION

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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